TITLE: Phan Tuan Kiệt 2552654 Date: 4/10/4

Activity 1.

~~have~~ $p_h = \frac{9}{12} =$

Let p_h is the probability of head in data set.

$\Rightarrow L(p_h) \propto p_h^9 (1-p_h)^{12-9}$

if the number of head in observed set increase which mean that the probability of head will increase to maximize the value of likelihood and vice versa.

Activity 2:

Let $\theta = 1$

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1
1	0	1
0	1	1
0	0	0
1	1	0

$L(\theta) = P(A=1 \cap B=0 | \theta) \propto P(\theta) \cdot P(\theta | A=1) \cdot P(\theta | B=0)$

$P(\theta) = \frac{4}{8} = \frac{1}{2} \Rightarrow P(\bar{\theta}) = \frac{1}{2}$

$P(\theta | A=1) = \frac{3}{4} \Rightarrow P(\bar{\theta} | A=1) = \frac{1}{4}$

$P(\theta | B=0) = \frac{2}{4} = \frac{1}{2} \Rightarrow P(\bar{\theta} | B=0) = \frac{1}{2}$

$\Rightarrow L(\theta) \propto \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{16}$

while $L(\bar{\theta}) = P(A=1 \cap B=0 | \bar{\theta}) \propto \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{16}$

$P(\theta | A=1 \cap B=0) = \frac{3/16}{3/16 + 1/16} = 0,75$

$\Rightarrow$ with A=1 and B=0 according to the dataset we have 75% to turn that the output is 1.

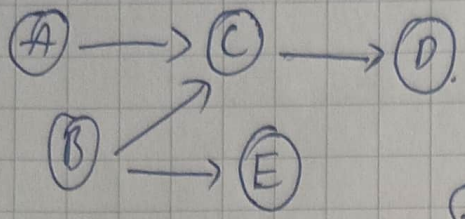
- for example in tosses coin, the probability of head in each tosses is 50%. But with the result ~~100 he~~ all head face in 100 times in a row (no affected by external interference) can be considered impossible.

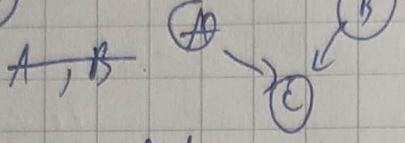
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Activity 3:



(i) $A \perp\!\!\!\perp B$  inactive

→ $A \perp\!\!\!\perp B$ be blocked.

(ii) $A \perp\!\!\!\perp B \mid C$ $A \to C$, $B \to C$ know C ⇒ active

→ $A \perp\!\!\!\perp B \mid C$ active

(iii) $A \perp\!\!\!\perp D \mid C$ $A \to C \to D$

because know C so don't need + ⇒ inactive

→ $A \perp\!\!\!\perp D \mid C$ blocked.

(iv) $C \perp\!\!\!\perp E \mid B$ $B \to C$, $B \to E$

because know B so don't need C ⇒ inactive

→ $C \perp\!\!\!\perp E \mid B$ blocked.

Activity 4:

ex: rain → wet street → accident

rain is reason leading to the accident because the reason of accident is wet street and rain cause wet street.

ex: studying → good exam ← good cheat sheet

studying and good cheat sheet are the evidence to conclude the result good exam, look one of them cannot conclud that result.

in the second example if don't know good exam, we cannot know the student have a good cheat sheet or not although know clearly they study very hard because to get the good exam, have to get both conditions.

Just write it

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